

## Intro to Cryo-EM / Cryo-ET High Performance Computing Cluster

*The UW-Madison Cryo-EM Research Center and Midwest Center for Cryo Electron Tomography have a dedicated computing cluster resource, which can accelerate your data processing. New cluster users should apply for an account at:*

[https://explore.wisc.edu/cryoem\\_hpc\\_access\\_request](https://explore.wisc.edu/cryoem_hpc_access_request)

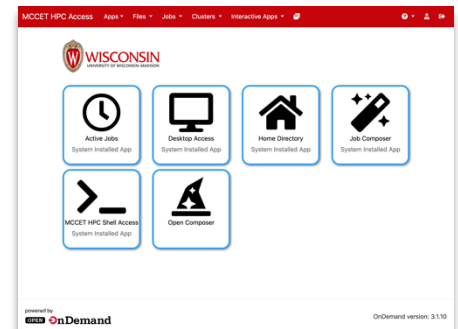
## Connecting to the Cryo-EM HPC Cluster

Open OnDemand portal

<https://cryoemportalhpc.wisc.edu>

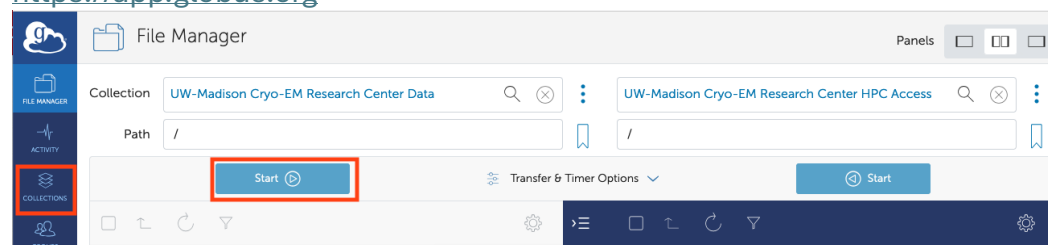
## Linux Commands

|                    |                                                             |
|--------------------|-------------------------------------------------------------|
| ls                 | Directory listing                                           |
| ls -l              | Formatted listing with information                          |
| cd <dir>           | Change to directory <dir>                                   |
| cd                 | Change to your home directory (/mnt/hpc_users/home/)        |
| pwd                | Show current directory                                      |
| mkdir <dir>        | Make a directory <dir>                                      |
| rm <file>          | Remove file                                                 |
| mv <file1> <file2> | Move a file to a new location                               |
|                    |                                                             |
| sinfo              | Give current status of the SLURM queue(s)                   |
| squeue             | List current submitted and running jobs.                    |
| scancel <id>       | Cancel a job id                                             |
| submit_to_a5000.sh | Submit a job to NVIDIA A5000 queue (requests node /w 4 GPU) |
| submit_to_a100.sh  | Submit a job to NVIDIA A100 queue (higher memory GPU cards) |
| submit_to_cpu.sh   | Submit a job to CPU-only systems, no GPU provided.          |



## Globus File Transfers

<https://app.globus.org>



Select your files in the collections you want to transfer, then choose “Start” to begin transferring data to your HPC user home directory, or bringing results back to own storage.

## Documentation available

Learn and get help using SLURM and the HPC cluster.

<https://uwcryoem.github.io/>



## Support and Help

Contact [jfscheuren@wisc.edu](mailto:jfscheuren@wisc.edu) and CC: [cryoem@biochem.wisc.edu](mailto:cryoem@biochem.wisc.edu) for email inquiries.

Attend the Office Hours on Thursdays at 9am at the Cryo-EM Office HF Deluca B1131 for questions and help using the HPC cluster.

## Common Scientific Software

|                                            |                                                                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Chimera and ChimeraX<br>IMOD, 3dmod, etomo | Viewer and visualizations for .mrc data and structures<br>Image processing, modeling and display programs used for tomographic reconstruction |
| IsoNet                                     | Reconstruct contents of the missing wedge in electron tomography                                                                              |
| RELION                                     | Single-particle analysis for Cryo-EM and Tomography processing workflow in Cryo-ET.                                                           |
| MotionCorr2 / MotionCorr3                  | Software packages for GPU accelerated motion correction of micrographs.                                                                       |
| AreTomo                                    | GPU-accelerated software for fiducial-free alignment and reconstruction for cryoEM tomography                                                 |
| Dynamo, emClarity, PEET<br>EMAN2           | Subtomogram averaging software packages.<br>Single-particle analysis for Cryo-EM and for cryo-ET image processing.                            |